

From Java Foundations to System Design & Production Engineering

This guide expands the journey from core Java programming to building large-scale, production-ready systems. It bridges the gap from language skills to system design and production engineering.

1. Java Foundations (Language Skills)

- 1 Data Structures & Algorithms
- 2 Concurrency & Multithreading
- 3 Networking & I/O (Blocking & Non-blocking)
- 4 Object-Oriented Programming & Design Principles

2. Spring Boot Deep Dive

- 1 Dependency Injection & Bean Lifecycle
- 2 Spring Data JPA & Transactions
- 3 Aspect-Oriented Programming
- 4 Spring MVC vs WebFlux

3. Microservices Architecture

- 1 Service decomposition and API design
- 2 Communication: REST, gRPC, Messaging
- 3 Patterns: API Gateway, Circuit Breaker, Saga

4. Cloud Engineering

- 1 Containers (Docker) and Orchestration (Kubernetes)
- 2 Cloud services: Compute, Storage, Databases
- 3 Scalability and fault tolerance patterns

5. Security Engineering

- 1 Authentication & Authorization (OAuth2, JWT)
- 2 Spring Security internals

- 3 API protection and threat mitigation

6. Observability

- 1 Logging, Monitoring, Metrics
- 2 Distributed tracing

7. Data & Performance Engineering

- 1 Caching (Redis), Search (Elasticsearch)
- 2 Event streaming (Kafka)
- 3 Performance tuning and scaling

8. System Design Thinking

- 1 Designing scalable distributed systems
- 2 CAP theorem and trade-offs
- 3 Real-world architectures (payments, chat, recommendations)

9. DevOps & CI/CD

- 1 Automation pipelines
- 2 Infrastructure as Code
- 3 Deployment strategies

10. AI-Enabled Backend Systems

- 1 RAG pipelines
- 2 Vector databases
- 3 AI microservices integration

Mastering this full stack transforms a developer into a system architect capable of designing scalable, secure, and intelligent platforms.